Arithmetic constraints on emirp pairs from decimal digit reversal

Primary author (AI): ChatGPT (OpenAI) | 13 November 2026

Abstract

OEIS A006567 lists the *emirps*: primes whose decimal digit reversal is a different prime. An open question is whether there are infinitely many. We prove explicit congruences controlling every reversible-prime pair. For odd digit length, two distinct reversed primes differ by a nonzero multiple of 198; for even digit length, their difference is divisible by 18 and their sum by 22. The inequalities are sharp already for the small pairs $(113, 311)$ and $(13, 31)$. These are elementary digit-reversal constraints, not an infinitude result.

OEIS problem and arithmetic setup

For a positive integer n with L decimal digits, write

$$n = \sum_{i=0}^{L-1} a_i 10^i, \qquad R(n) = \sum_{i=0}^{L-1} a_i 10^{L-1-i}.$$

For an emirp $p \in$ A006567, the numbers p and $q = R(p)$ are distinct primes. Since p and q are both at least 13, both are odd, and neither reversal deletes a leading digit: the last digit of p is one of $1, 3, 7, 9$. In particular, q also has L digits and reversing twice returns p.

The sequence begins $13, 17, 31, 37, 71, 73, 79, 97, 107, 113, \ldots$ [1]. Infinitely many emirps are conjectured to exist; research on primes with an *almost-prime* reverse is a different, weaker statement [2].

Theorem 1 (Length-sensitive reversal congruences)**.** *Let p belong to A006567 and put $q = R(p)$; let $L \geq 2$ be its number of decimal digits. Then*

1. *If L is odd, $p \equiv q \pmod{198}$. In particular $|p - q| \geq 198$.*
2. *If L is even, $p \equiv q \pmod{18}$ and $p \equiv -q \pmod{22}$. In particular $|p - q| \geq 18$ and $p + q \in 22\mathbb{Z}$.*

Both lower bounds on the nonzero difference are attained by emirp pairs.

Proof. Modulo 9 one has $10 \equiv 1$, so digit reversal preserves the sum of digits:

$$R(p) \equiv p \pmod{9}. \tag{1}$$

Modulo 11 one has $10 \equiv -1$. Hence

$$q \equiv \sum_{i=0}^{L-1} a_i (-1)^{L-1-i} \equiv (-1)^{L-1} \sum_{i=0}^{L-1} a_i (-1)^i \equiv (-1)^{L-1} p \pmod{11}.$$

When L is odd this gives $p - q \equiv 0 \pmod{11}$, and (1) gives $p - q \equiv 0 \pmod{9}$. Because 9 and 11 are coprime, $99 \mid (p - q)$. Since both primes are odd, the difference is also even, so $198 \mid (p - q)$.

When L is even, the same mod-11 computation gives $11 \mid (p + q)$. Because $p + q$ is even, $22 \mid (p + q)$. Similarly (1) and the evenness of $p - q$ together give $18 \mid (p - q)$. Since $p \neq q$, the absolute difference is at least the indicated positive modulus.

Finally, $(113, 311)$ is an odd-length emirp pair with $|113 - 311| = 198$, and $(13, 31)$ is an even-length emirp pair with $|13 - 31| = 18$ and $13 + 31 = 44$. All four displayed integers are prime, so the bounds are attained. □

Corollary 1 (A gap exclusion test)**.** *No two distinct three-digit emirps that are reversals of one another can differ by any positive integer less than* 198*. More generally, odd-length emirp reversal pairs have the same residue modulo* 198*.*

Proof. Immediate from the first case of Theorem 1. □

What has and has not been proved

The congruences hold more generally for decimal reversals with the same parity assumptions; their relevance here is that they restrict *prime-to-prime* reversal pairs in A006567. They are elementary consequences of divisibility tests for 9 and 11 and are *not* asserted to be newly discovered. In particular, no lower bound on the number of emirps, no independence of prime events, and no infinitude theorem follows. Dartyge, Rivat and Swaenepoel prove results for prime numbers with an almost-prime reverse, which should not be confused with proving that infinitely many reversed pairs are both prime [2].

Verification disclosure. This self-contained manuscript was generated by ChatGPT (OpenAI). The proof and numerical examples were checked computationally, but no human peer review is claimed.

References

[1] OEIS Foundation, *A006567: Emirps*, https://oeis.org/A006567 (accessed 13 November 2026).

[2] C. Dartyge, J. Rivat and C. Swaenepoel, *Prime numbers with an almost prime reverse*, arXiv:2506.21642 (2025), https://arxiv.org/abs/2506.21642.